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Batch: 15

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Experiment-39

1. Variadic Function — Weighted Average

Algorithm

1. Start.
2. Define the function `weightedAverage(int count, ...)`.
3. Initialize `weightedSum = 0` and `totalWeight = 0`.
4. Start processing variable arguments using `va_start()`.
5. Repeat `count` times:
 - o Read `value`.
 - o Read `weight`.
 - o If `weight` is negative, display an error and return.
 - o Calculate `value * weight` and add it to `weightedSum`.
 - o Add `weight` to `totalWeight`.
6. End variable argument processing using `va_end()`.
7. If `totalWeight` is 0, display an error.
8. Otherwise calculate weighted average as `weightedSum / totalWeight`.
9. Return the weighted average.
10. Display the result in `main()`.
11. Stop.

C Program

```
#include <stdio.h>

#include <stdarg.h>

double weightedAverage(int count, ...)
{
    va_list args;

    double value, weight;

    double weightedSum = 0.0;

    double totalWeight = 0.0;
```

```
va_start(args, count);
```

```
for (int i = 0; i < count; i++)
```

```
{
```

```
    value = va_arg(args, double);
```

```
    weight = va_arg(args, double);
```

```
    if (weight < 0)
```

```
    {
```

```
        printf("Error: Weight cannot be negative.\n");
```

```
        va_end(args);
```

```
        return -1;
```

```
    }
```

```
    weightedSum = weightedSum + (value * weight);
```

```
    totalWeight = totalWeight + weight;
```

```
}
```

```
va_end(args);
```

```
if (totalWeight == 0)
```

```
{
```

```
    printf("Error: Total weight cannot be zero.\n");
```

```
    return -1;
```

```
}
```

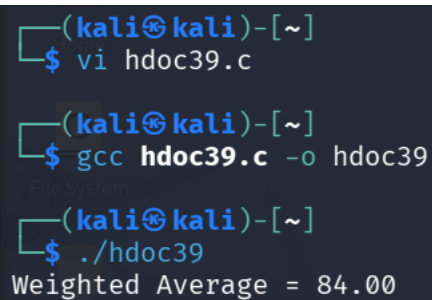
```
return weightedSum / totalWeight;
```

```
}
```

```
int main()
```

```
{  
    double result;  
  
    result = weightedAverage(3,  
                             80.0, 0.3,  
                             75.0, 0.2,  
                             90.0, 0.5);  
  
    if (result != -1)  
        printf("Weighted Average = %.2f\n", result);  
  
    return 0;  
}
```

Compilation And Execution



```
(kali㉿kali)-[~]  
$ vi hdoc39.c  
  
(kali㉿kali)-[~]  
$ gcc hdoc39.c -o hdoc39  
  
(kali㉿kali)-[~]  
$ ./hdoc39  
Weighted Average = 84.00
```